

1. What is Big O Notation?

Big O notation is a mathematical notation used to describe the efficiency of an algorithm by measuring how its execution time or memory usage grows as the input size increases.

2. Search Cases

Case	Linear Search	Binary Search
Best Case	$O(1)$	$O(1)$
Average Case	$O(n)$	$O(\log n)$
Worst Case	$O(n)$	$O(\log n)$

3. Time Complexity Comparison

Algorithm	Time Complexity
Linear Search	$O(n)$
Binary Search	$O(\log n)$

4. Which is Better?

Binary Search is more suitable for an e-commerce platform because:

- It searches much faster than linear search.
- It is efficient for large datasets.
- It requires the product array to be sorted before searching.